EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

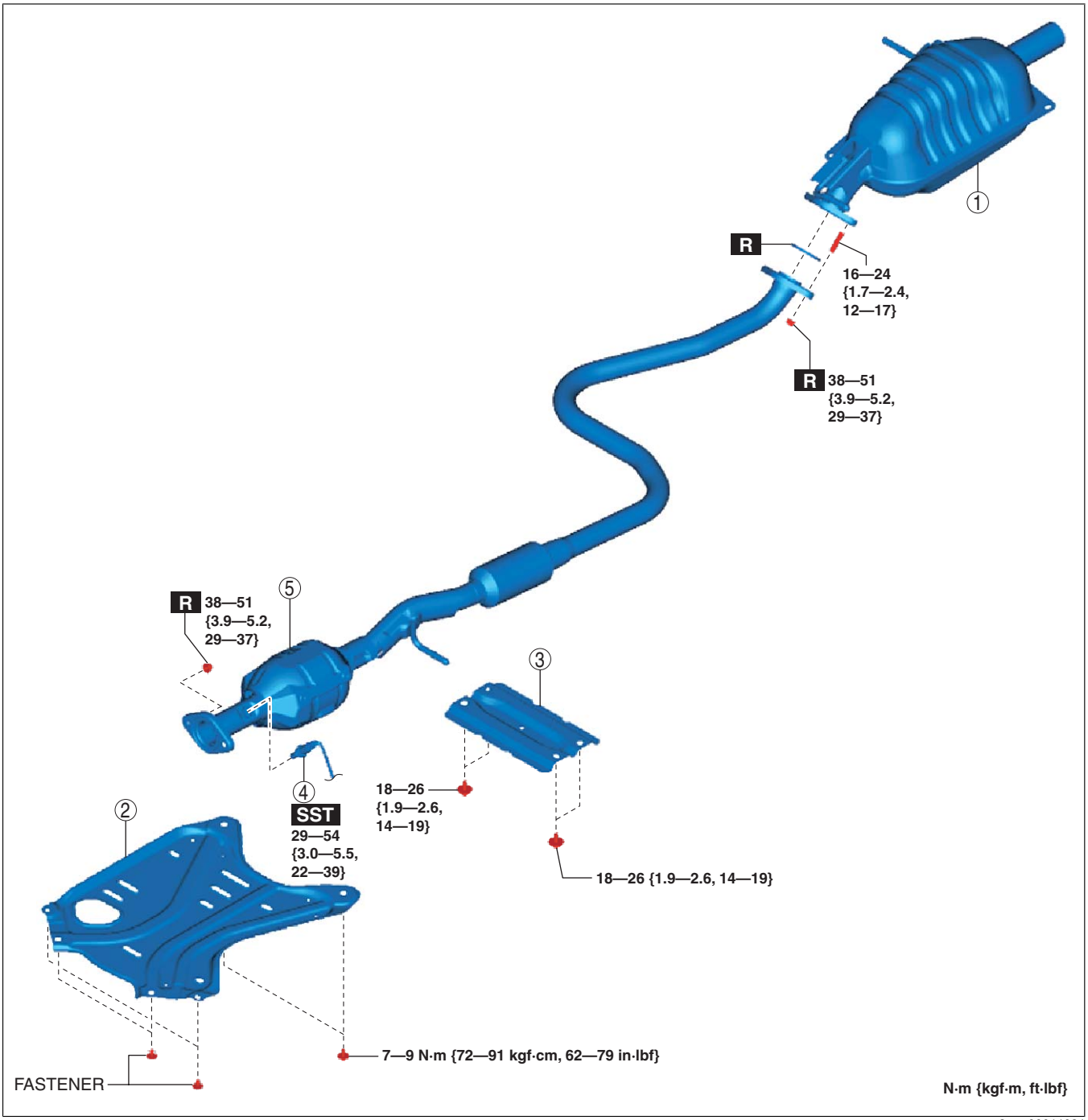
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Warning

- **A hot engine and exhaust system can cause severe burns. Turn off the engine and wait until they are cool before removing the exhaust system.**

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the plug hole plate. (See PLUG HOLE PLATE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
3. Remove in the order indicated in the table.
4. Remove the exhaust system insulator. (See Exhaust System Insulator Removal/Installation.)
5. Install in the reverse order of removal.

Step 1



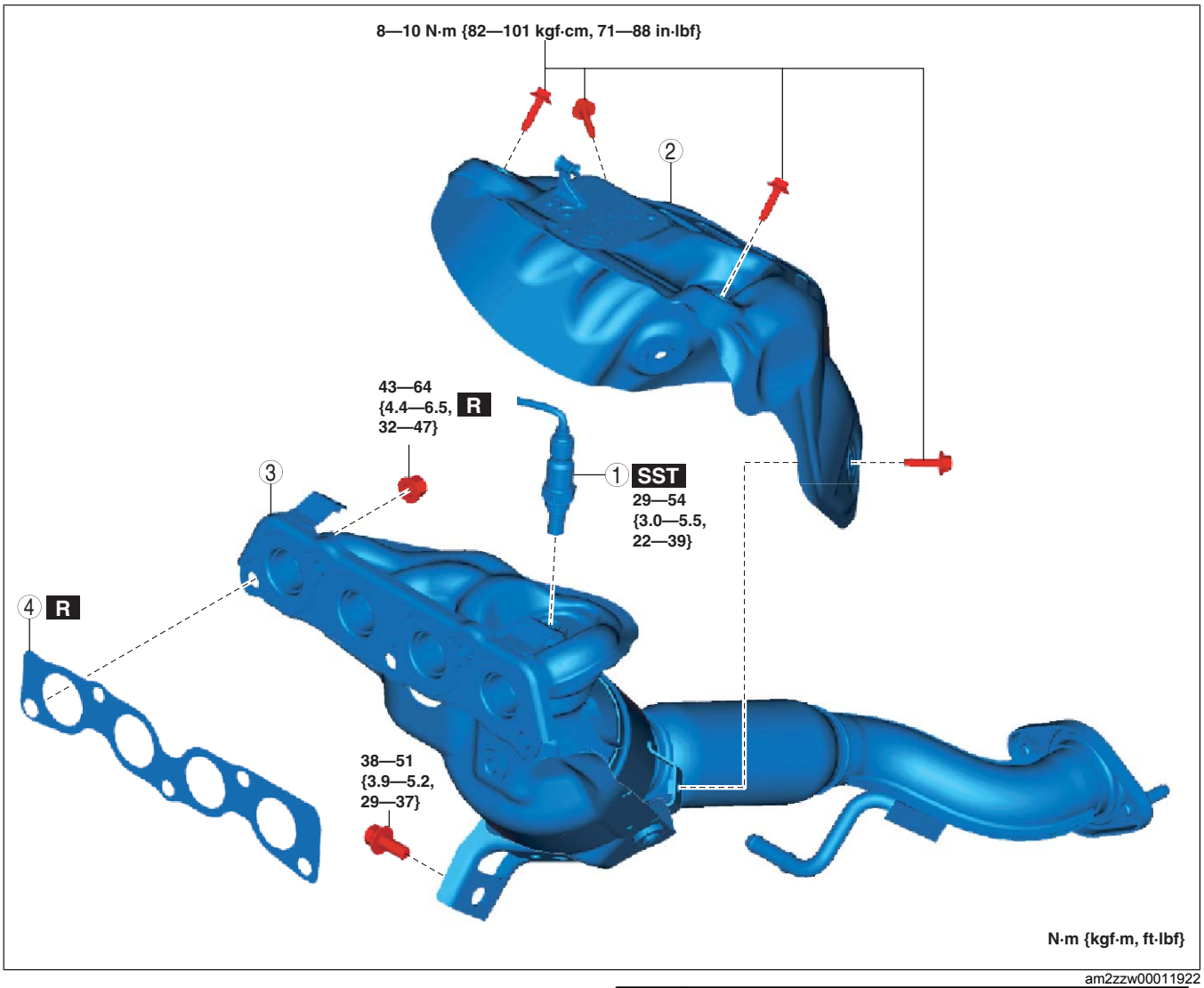
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1	Main silencer
2	Insulator
3	Tunnel member

4	HO2S (See HEATED OXYGEN SENSOR (HO2S) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

5	TWC

Step 2 (SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system))

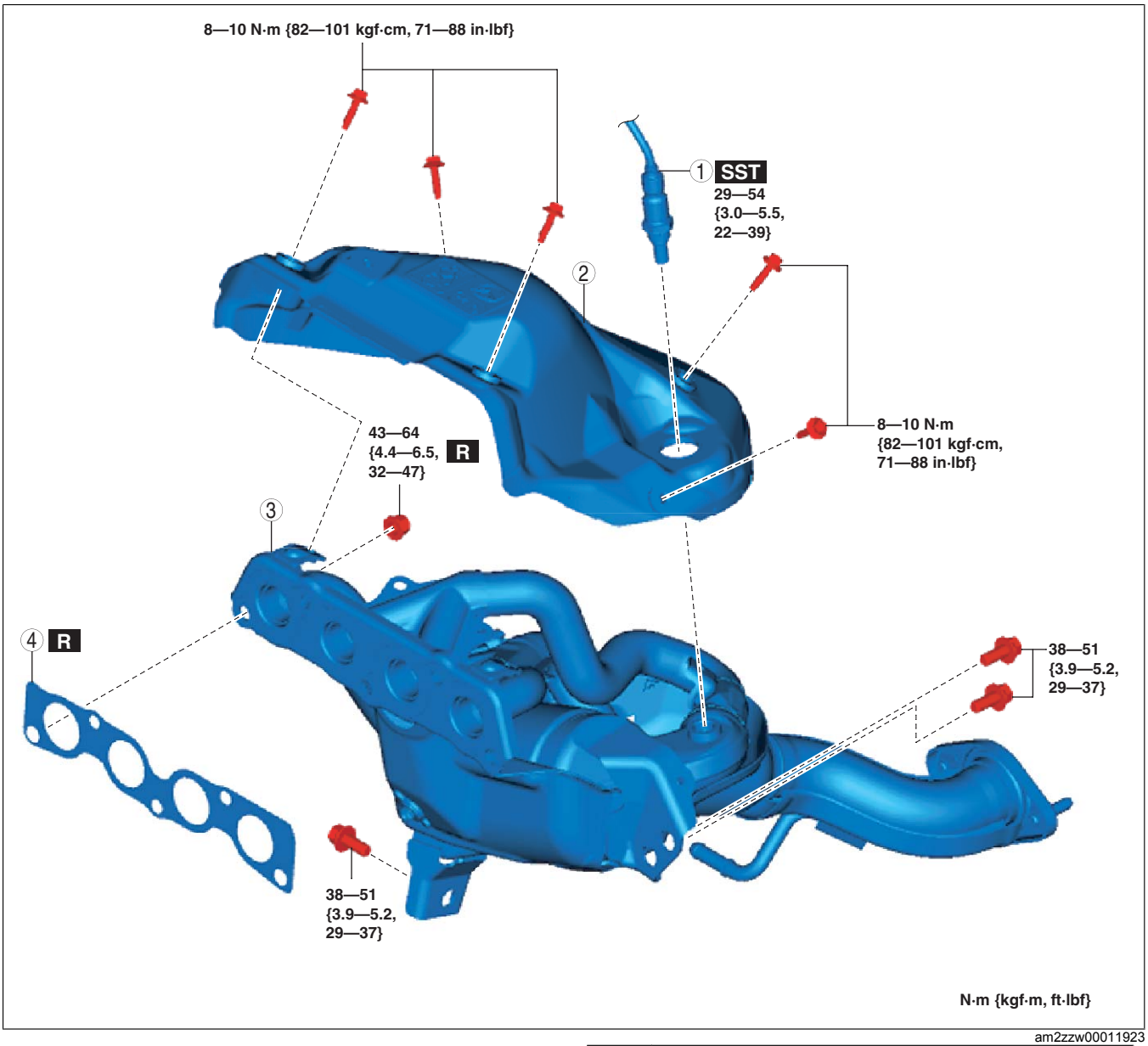


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1	A/F sensor (See AIR FUEL RATIO (A/F) SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
2	Exhaust manifold insulator (See Exhaust Manifold Insulator Installation Note.)

3	Exhaust manifold (WU-TWC) (See Exhaust Manifold Removal Note.) (See Exhaust Manifold Installation Note.)
4	Exhaust manifold gasket

Step 2 (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))



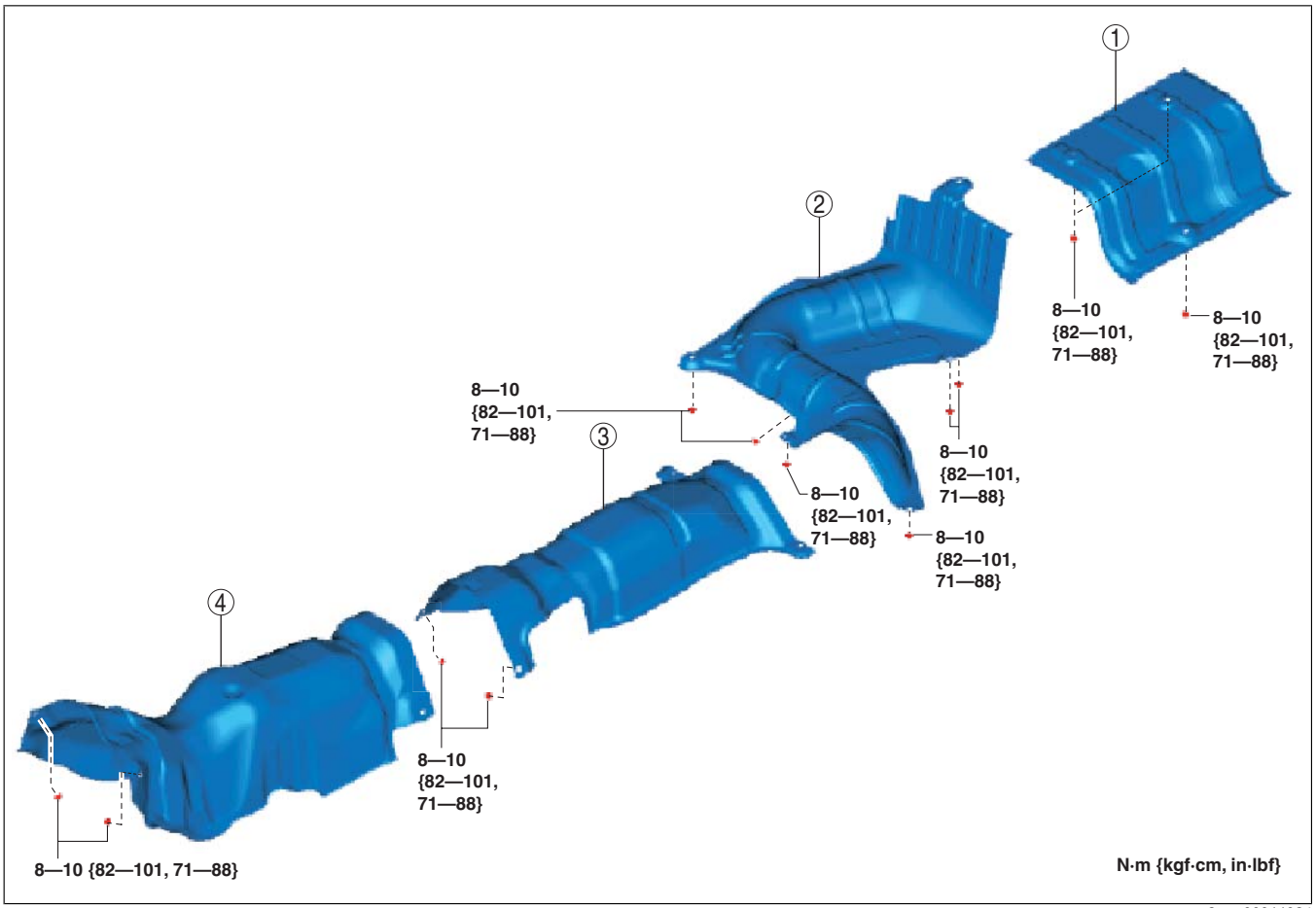
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1	A/F sensor (See AIR FUEL RATIO (A/F) SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
2	Exhaust manifold insulator (See Exhaust Manifold Insulator Removal Note.) (See Exhaust Manifold Insulator Installation Note.)

3	Exhaust manifold (WU-TWC) (See Exhaust Manifold Removal Note.) (See Exhaust Manifold Installation Note.)
4	Exhaust manifold gasket

Exhaust System Insulator Removal/Installation

1. Remove the exhaust system insulator in the order shown in the figure.
2. Install in the reverse order of removal.



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1	Insulator (rear)
2	Insulator (middle No.1)
3	Insulator (middle No.2)
4	Insulator (front)

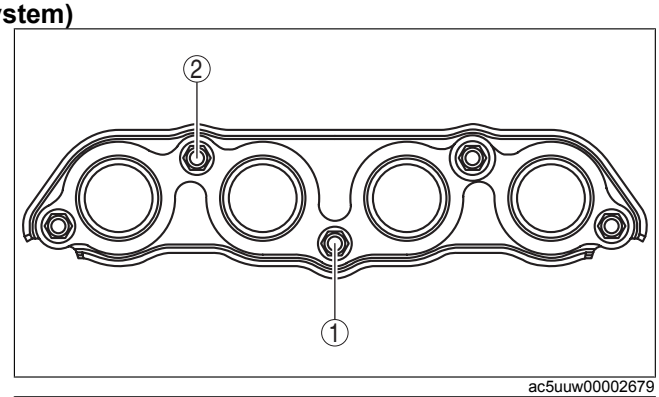
Exhaust Manifold Insulator Removal Note
SKYACTIV-G 1.5 (with 4-2-1 exhaust system)
1. Remove the windshield wiper arm and blade. (See WINDSHIELD WIPER ARM AND BLADE REMOVAL/INSTALLATION.)
2. Remove the cowl grille. (See COWL GRILLE REMOVAL/INSTALLATION.)
3. Remove the exhaust manifold insulator.

Exhaust Manifold Removal Note
SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)
1. Remove the front under cover No.2. (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
2. Remove the windshield wiper arm and blade. (See WINDSHIELD WIPER ARM AND BLADE REMOVAL/INSTALLATION.)
3. Remove the cowl grille. (See COWL GRILLE REMOVAL/INSTALLATION.)
4. Remove the windshield wiper motor. (See WINDSHIELD WIPER MOTOR AND LINK REMOVAL/INSTALLATION.)
5. Remove the cowl panel. (See COWL PANEL REMOVAL/INSTALLATION.)
6. Remove the exhaust manifold.

SKYACTIV-G 1.5 (with 4-2-1 exhaust system)
1. Remove the front crossmember. (See FRONT CROSSMEMBER REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
2. Remove the exhaust manifold.

Exhaust Manifold Installation Note
SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)

1. Temporarily tighten the exhaust manifold installation nuts (1) and (2) shown in the figure by hand.
2. Tighten the exhaust manifold installation nuts (1) and (2) shown in the figure to **33—53 N·m {3.4—5.4 kgf·m, 25—39 ft·lbf}**.



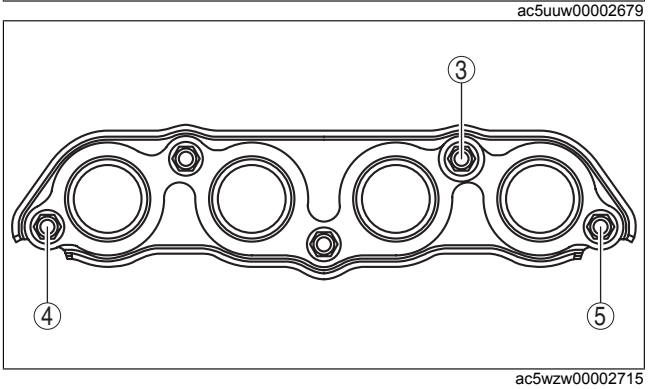
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3. Temporarily tighten the exhaust manifold installation nuts (3) to (5) shown in the figure by hand.
4. Tighten the exhaust manifold installation nuts (3) to (5) shown in the figure.

   Exhaust manifold installation nut tightening torque
   43—64 N·m {4.4—6.5 kgf·m, 32—47 ft·lbf}

5. Tighten the exhaust manifold installation nuts (1) and (2) shown in the figure.

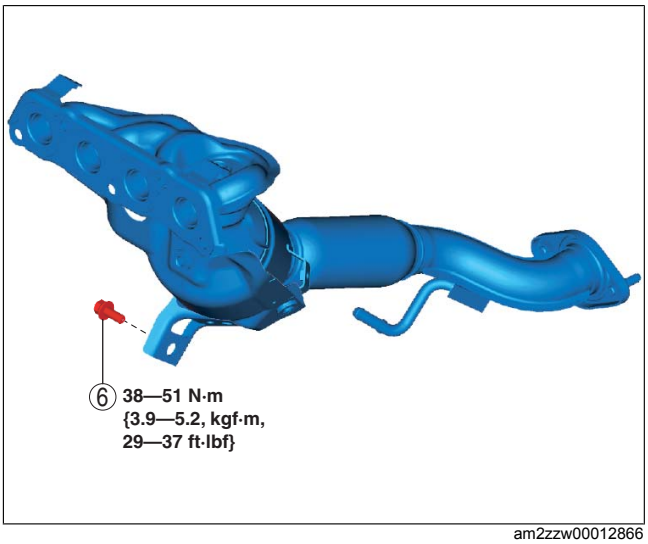
   Exhaust manifold installation nut tightening torque
   43—64 N·m {4.4—6.5 kgf·m, 32—47 ft·lbf}



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6. Temporarily tighten the bolt (6) shown in the figure.
7. Tighten the bolt (6) shown in the figure.

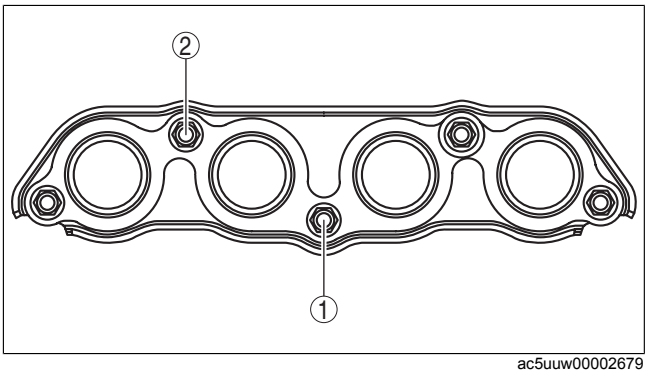
   Exhaust manifold installation bolt tightening torque
   38—51 N·m {3.9—5.2 kgf·m, 29—37 ft·lbf}



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SKYACTIV-G 1.5 (with 4-2-1 exhaust system)

1. Temporarily tighten the exhaust manifold installation nuts (1) and (2) shown in the figure by hand.
2. Tighten the exhaust manifold installation nuts (1) and (2) shown in the figure to **33—53 N·m {3.4—5.4 kgf·m, 25—39 ft·lbf}**.



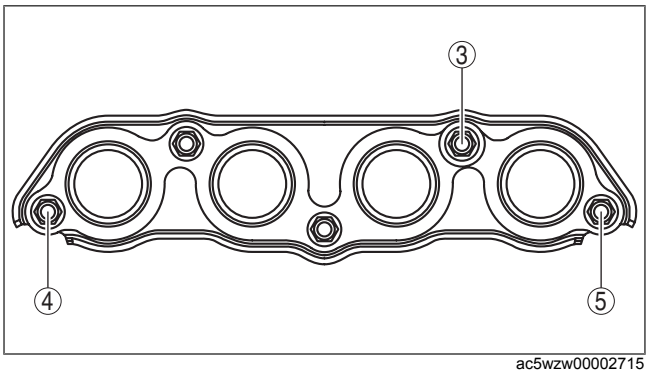
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3. Temporarily tighten the exhaust manifold installation nuts (3) to (5) shown in the figure by hand.
4. Tighten the exhaust manifold installation nuts (3) to (5) shown in the figure.

   Exhaust manifold installation nut tightening torque
   43—64 N·m {4.4—6.5 kgf·m, 32—47 ft·lbf}

5. Tighten the exhaust manifold installation nuts (1) and (2) shown in the figure.

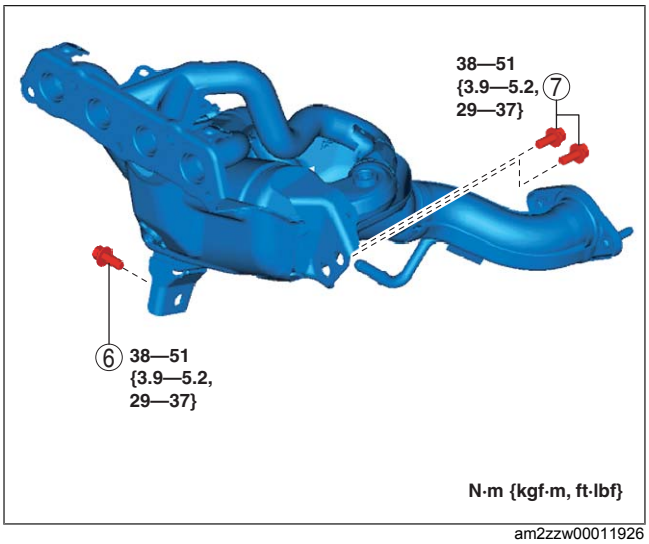
   Exhaust manifold installation nut tightening torque
   43—64 N·m {4.4—6.5 kgf·m, 32—47 ft·lbf}



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6. Temporarily tighten the bolts (6) and (7) shown in the figure.
7. Tighten the bolts (6) and (7) shown in the figure.

   Exhaust manifold installation bolt tightening torque
   38—51 N·m {3.9—5.2 kgf·m, 29—37 ft·lbf}



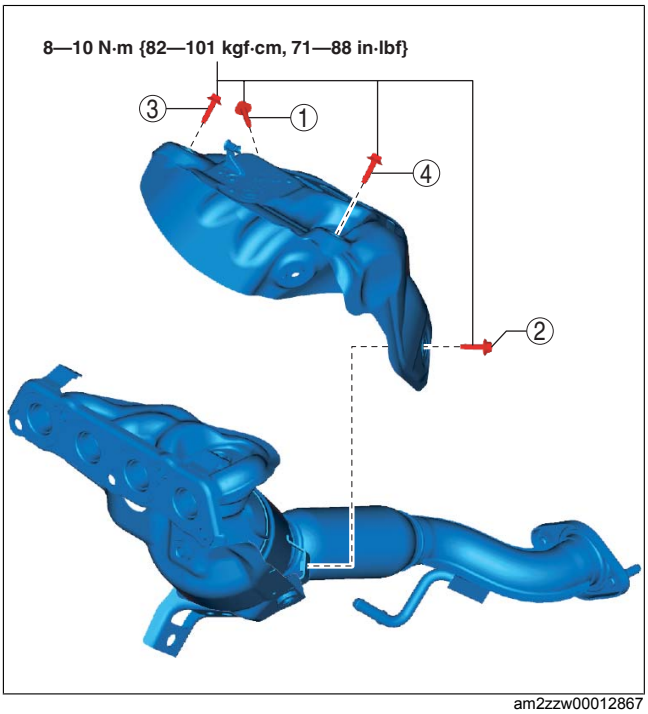
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Exhaust Manifold Insulator Installation Note

Caution
- **Since the quality of the material of exhaust manifold insulator is aluminum material, it is very easy to change. Therefore, it takes care that there is no modification of exhaust manifold insulator at the time of attachment for dealing with it. (SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system))**

1. Temporarily tighten the exhaust manifold insulator.
2. Tighten the exhaust manifold insulator in the order shown in the figure.

SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)

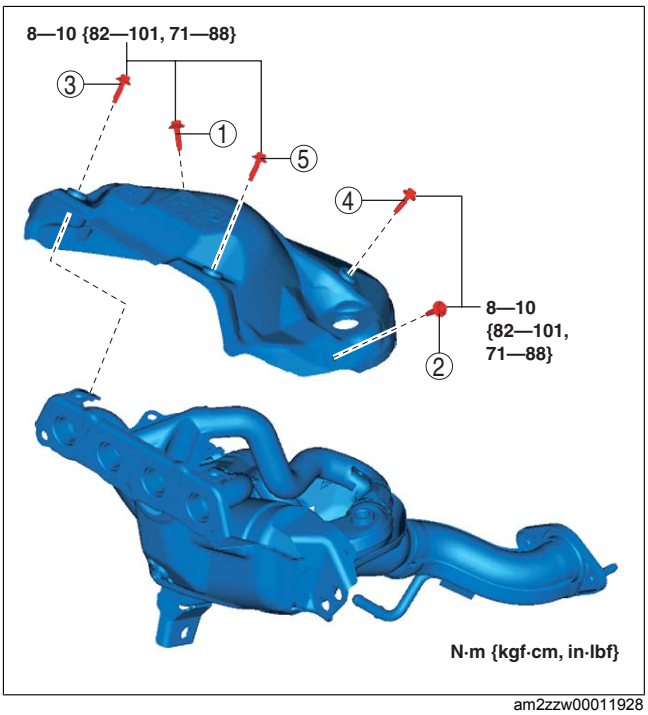


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SKYACTIV-G 1.5 (with 4-2-1 exhaust system)

Exhaust manifold insulator tightening torque
8—10 N·m {82—101 kgf·cm, 71—88 in·lbf}

3. Check that there is no contact with an exhaust manifold and exhaust manifold insulator.



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